

RS001 Week 14 — English Worksheet

Topic: Random Encounters + a Bigger Reward Pool · **Course:** Text Adventure (Python) · **Time:** about 45 minutes

This week is about thinking through randomness and explaining it in your own words. Each path draws a random enemy from its own pool, rewards come from a big pool, and a surprise encounter fires only **some** of the time using a weighted **True** / **False** pick. You will read code, fix a broken version on paper, then explain the game you wrote in your live lesson.

🗨 Words to know: **random**, **pool**, **encounter**, **odds**, **different**

```
import random
left_enemies = ["goblin", "bat", "spider"]
enemy = random.choice(left_enemies)

if random.choice([True, False, False]): # about 1 in 3
    game.say("A bandit jumps out!")
```

1 · Predict 🧠

Read each piece of code. Run it in your head and write what you think happens.

```
import random
left_enemies = ["goblin", "bat", "spider"]
enemy = random.choice(left_enemies)
game.say(f"You meet a {enemy}.")
```

Can you say for sure which enemy appears? Why or why not?


```
import random
if random.choice([True, False, False]):
    game.say("ambush!")
else:
    game.say("quiet path")
```

Out of the three items, how many are **True**? About how often does "ambush!" show?


```
import random
if random.choice([True, True, True, False]):
    game.say("rare event")
```

How does changing the mix of True/False change how often the event happens?

2 · Spot the Bug 🐛

Each block was meant to do something, but it is broken. Read what it is **supposed** to do, fix it on paper, then explain why the original was wrong.

Bug A — Each path should draw from its **own** enemy pool. Right now both paths draw from the same list, so left and right feel identical.

```
import random
left_enemies = ["goblin", "bat"]
right_enemies = ["orc", "troll"]
path = "right"
if path == "left":
    enemy = random.choice(left_enemies)
else:
    enemy = random.choice(left_enemies)
game.say(enemy)
```

Hint: the right path should pull from `right_enemies`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A rare event should fire only about 1 in 3 times. Right now the mix makes it fire almost every time.

```
import random
if random.choice([True, True, False]):
    game.say("A rare treasure glows nearby!")
```

Hint: to make `True` rarer, change how many `True` vs `False` are in the list.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The reward pool should hold several rewards so the pick feels varied. Right now the list has only one item, so it is the "same random" every time.

```
import random
rewards = ["gold"]
reward = random.choice(rewards)
game.say(f"You found {reward}")
```

Hint: a one-item pool always returns that one item — give it more choices.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working example. It picks a random enemy from the path's own pool, draws a reward from a bigger pool, and fires a surprise encounter only some of the time.

```
import random

left_enemies = ["goblin", "bat", "spider"]
right_enemies = ["orc", "troll", "wolf"]
rewards = ["gold", "potion", "map", "gem", "key"]

path = "left"
if path == "left":
    enemy = random.choice(left_enemies)
else:
    enemy = random.choice(right_enemies)
game.say(f"A {enemy} blocks the path!")

reward = random.choice(rewards)
game.say(f"You won a {reward}.")

if random.choice([True, False, False]): # about 1 in 3
    game.say("A bandit jumps out!")
```

Which list does the enemy come from when `path` is `"left"` ?

Why does the `reward` line give a different result on different playthroughs?

Look at `[True, False, False]` . About how often does the bandit appear?

If you wanted the bandit to appear more often, how would you change that list?

In your own words, why does this code make every playthrough feel different?

4 · Explain Your Lesson Code 🎥

Take a short video on your phone (or a parent's phone) and explain the game **you** wrote in today's live lesson. You can show it running. Teach it like the viewer has never coded. Try to use these words: **random, pool, encounter, odds, different**.

1. Point to your enemy pools and say what is in each path's pool.
2. Show your reward pool and play once so a random reward appears.
3. Point to your weighted `True/False` line and explain the odds of the surprise encounter.
4. Say why every playthrough now feels different.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.